

RESEARCH PAPER

Product Line Profitability & Margin Performance Analysis

Nassau Candy Distributor

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Tools Used: Python | MySQL | Microsoft Excel | Power BI | Streamlit

1. Abstract

This research paper documents a complete end-to-end business analytics project conducted on Nassau Candy Distributor's transaction dataset as part of the Unified Mentor Data Analytics Internship Program. As a recent BSc Data Analytics graduate undertaking my first full industry project, I approached this analysis by first understanding the business context before touching any tool — a discipline that shaped every analytical decision made throughout the project.

The dataset comprised 10,194 transaction records spanning 15 products across 3 divisions and 4 geographic regions. The analysis was structured around four core business questions provided by the organisation: identifying which product lines deliver the highest gross margin, whether high-sales products are genuinely profitable, how profitability varies across divisions, and which products represent margin risk. Each question was answered using a multi-tool pipeline — Python for data preparation and analysis, MySQL for structured business queries, Microsoft Excel for pivot-based business summaries, Power BI for interactive dashboards, and Streamlit for live web-based analytics. The most striking finding was that a single division — Chocolate — accounts for 92.88% of total revenue and 95% of total profit, revealing a level of business concentration that represents both strength and significant risk.

2. Introduction

2.1 About Nassau Candy Distributor

Nassau Candy is a US-based specialty candy and food distributor operating across four geographic regions: Interior, Atlantic, Gulf, and Pacific. As a distributor operating between manufacturers and retailers, the company's profitability is directly tied to its ability to manage margins across a diverse product portfolio. The company's 15 products are organised into three divisions: Chocolate, Sugar, and Other.

2.2 Problem Statement

The organisation currently lacks visibility into the following four critical business questions:

- Which product lines deliver the highest gross margin
- Whether high-sales products are actually profitable
- How profitability varies across product divisions
- Which products represent margin risk

2.3 My Approach to This Project

When I first received this dataset, my instinct was not to open a tool immediately. Instead, I spent time understanding what Nassau Candy actually does as a business — a distributor squeezed between manufacturers and retailers — before defining what metrics I needed to calculate. After reviewing the 18 columns in the dataset, I made a deliberate decision about tool sequencing: Python and SQL would carry the heaviest analytical workload since the dataset required significant computation and structured querying, while Excel, Power BI, and Streamlit would handle the presentation and business-readable output layers. This decision was driven by looking at the data first, not by habit.

This project was my first complete end-to-end analytics project, and I consciously chose to work through logical errors and syntax mistakes independently before seeking guidance — a discipline that significantly deepened my understanding of each tool.

3. Analytical Methodology

3.1 Tool Selection and Justification

After examining the dataset structure, I selected tools based on what each genuinely does best — not simply to demonstrate tool breadth, but because each tool served a specific purpose in the analytical pipeline:

Tool	Role	Why This Tool at This Stage
Python (pandas)	Data cleaning, metric derivation, EDA, visualisation	Row-level computation and derived column creation is most natural in pandas. All core metrics were derived here first.
MySQL	Structured business queries, cross-validation	Simulates real-world database environments. Used to independently validate Python findings through SQL GROUP BY and HAVING queries.
Microsoft Excel	Pivot Table analysis, business summaries	Non-technical business stakeholders can read Excel output without any tool knowledge. Pivot Tables confirmed findings in a third independent environment.
Power BI	Interactive visual dashboard	Enables drill-down filtering by Division and Region — richer interactivity than static charts.
Streamlit	Live web analytics dashboard	Python-based app that makes the analysis accessible via browser to anyone without requiring tool installation.

3.2 Data Cleaning and Validation

The first analytical discipline I applied was to diagnose the data before treating it — similar to a doctor examining a patient before prescribing medicine. The following validation checks were performed programmatically in Python:

- Shape confirmed: 10,194 rows, 18 columns in raw dataset (21 columns after derived metrics)
- Missing values: 0 null records across all columns — no imputation required
- Duplicate records: 0 duplicates found — all 10,194 rows are unique transactions
- Zero or negative Sales: 0 invalid records confirmed
- Zero or negative Gross Profit: 0 invalid records confirmed
- Data types: Sales, Cost, and Gross Profit loaded as object type in one environment — converted to float64 using `pd.to_numeric()` before analysis
- 15 unique products confirmed across 3 divisions and 4 regions

3.3 Derived Metrics

The raw dataset contained Sales, Cost, and Gross Profit columns but no margin or efficiency metrics. I derived the following four columns to enable the profitability analysis:

- Gross Margin % = (Gross Profit / Sales) x 100 — the central metric for the entire project, measuring how much of each sales dollar is retained as profit
- Profit per Unit = Gross Profit / Units — measures return on each individual unit sold
- Profit Contribution % = (Product Gross Profit / Total Gross Profit) x 100 — measures each product's share of the total profit pool
- Risk Flag = Categorical classification based on Gross Margin % thresholds: Healthy (above 60%), Reprice (40-60%), Discontinue Review (below 40%). Thresholds were set relative to the dataset's own average margin rather than arbitrary external benchmarks.

3.4 Cross-Validation Across Tools

A key discipline I maintained throughout this project was verifying that the same analysis performed in Python, SQL, and Excel produced identical results. For example, the division-level average Gross Margin % calculation was independently computed in all three tools:

- Python groupby result: Chocolate 67.46%, Other 37.67%, Sugar 57.69%
- MySQL GROUP BY result: Chocolate 67.47%, Other 37.51%, Sugar 57.69%
- Excel Pivot Table result: Chocolate 67.46%, Other 37.67%, Sugar 57.69%

Minor floating point differences between Python and MySQL are expected due to double precision handling. The consistency across tools confirmed the integrity of the analysis.

4. Exploratory Data Analysis

4.1 Dataset Overview

Parameter	Value
Total Transactions	10,194
Total Unique Products	15
Product Divisions	3 (Chocolate, Sugar, Other)
Geographic Regions	4 (Interior, Atlantic, Gulf, Pacific)
Total Sales Revenue	\$141,783.63
Total Gross Profit	\$93,442.80
Overall Average Gross Margin %	66.51%
Missing Values	0
Duplicate Records	0
Date Range	2024 transaction data

4.2 Initial Observations

My first analytical step after loading the data was to look at it — not run functions on it. Using `head()`, `describe()`, and `value_counts()`, I built a mental model of the dataset before writing any analysis code. Key initial observations included:

- The dataset is transaction-level — one row represents one order line, not one product
- Sales values range from \$1.25 to \$260.00 with a mean of \$13.91, suggesting a wide price range across products
- Gross Profit values range from \$0.25 to \$130.00 — with a mean of \$9.17
- Units per transaction range from 1 to 14, with a median of 3 units
- The top selling product by order count was Wonka Bar - Triple Dazzle Caramel at 7,596 orders; the least ordered product was Fun Dip at 8 orders — a striking 949x difference in order frequency

5. Key Findings

5.1 Finding 1 — Which Product Lines Deliver the Highest Gross Margin?

Division	Avg Gross Margin %	Total Sales	Total Gross Profit	Assessment
Chocolate	67.46%	\$131,692.90	\$88,824.62	Strongest — both margin and volume
Sugar	57.69%	\$427.48	\$284.73	Decent margin, negligible volume
Other	37.67%	\$9,663.25	\$4,333.45	Structural margin problem

Chocolate is the clear answer to the first business question — it leads not just in gross margin percentage but in total revenue and total profit by an overwhelming margin. When I first saw this result, my reaction was: this company is hugely dependent on Chocolate. That dependency became the thread connecting every subsequent finding in this analysis.

5.2 Finding 2 — Are High-Sales Products Actually Profitable?

This was the finding I found most analytically interesting. After grouping by Product Name and sorting by total Sales, I compared each product's sales rank against its margin rank:

Product	Sales Rank	Total Sales	Gross Margin %	Margin Rank
Wonka Bar - Triple Dazzle	1st	\$28,485	65.33%	4th

Caramel				
Wonka Bar - Scrumdiddlyumptious	2nd	\$27,875	69.44%	2nd
Wonka Bar - Milk Chocolate	3rd	\$26,868	64.92%	5th
Wonka Bar - Fudge Mallows	4th	\$24,890	66.67%	3rd
Wonka Bar - Nutty Crunch Surprise	5th	\$23,575	71.35%	1st

The answer is: not always. The highest selling product (Triple Dazzle Caramel) is only the 4th most margin-efficient. The lowest selling among the top 5 (Nutty Crunch Surprise) actually delivers the best margin. This confirmed the business question's premise — sales volume alone is a misleading measure of product value.

I identified Wonka Bar - Milk Chocolate as the primary margin risk product: it had the highest unit volume at 8,267 units AND the lowest gross margin among top performers at 64.92%. Both factors together — high volume with below-average margin — is what flagged it as a concern. Nassau Candy is processing more transactions for this product than any other while earning the least per sale.

5.3 Finding 3 — How Does Profitability Vary Across Divisions?

The division-level analysis revealed the most dramatic finding in the entire project. The gap between Chocolate (67.46%) and Other (37.67%) is nearly 30 percentage points — a structural difference, not a marginal one. More significantly, when looking at total sales contribution, Chocolate accounts for 92.88% of all revenue while Other accounts for 6.82% and Sugar just 0.30%.

This means Nassau Candy is operating what is effectively a single-division business. The Other and Sugar divisions exist in the portfolio but contribute almost nothing to the company's financial performance. This was not a conclusion I expected when I started the analysis — I anticipated more balance across divisions.

5.4 Finding 4 — Pareto Analysis: Profit Concentration Risk

I conducted a cumulative profit contribution analysis to understand how concentrated Nassau Candy's profit base is:

- Top 1 product: 20.7% of total profit
- Top 2 products: 40.6% of total profit
- Top 4 products: 77.3% of total profit
- Top 5 products: 95.1% of total profit
- Remaining 10 products: only 4.9% of total profit

Just 5 Wonka Bar variants — representing 33% of the product catalogue — generate 95% of all profit. This level of concentration means that any supply disruption, price pressure, or demand shift affecting a single Wonka Bar product could materially damage the entire company's profitability. For a distributor with no manufacturing control, this is a significant strategic vulnerability.

5.5 Finding 5 — Regional Performance

Region	Total Sales	Avg Gross Margin %	Sales Rank	Margin Rank
Pacific	\$46,301.53	66.59%	1st	2nd
Atlantic	\$41,197.24	66.21%	2nd	4th
Interior	\$32,037.60	66.87%	3rd	1st
Gulf	\$22,247.26	66.42%	4th	3rd

Regional margin differences are relatively small — all four regions sit within a 0.66 percentage point range (66.21% to 66.87%). However, the sales rank and margin rank do not align: Pacific leads in sales but Interior leads in margin efficiency. Atlantic is second in revenue but last in margin, suggesting potential pricing or product mix differences in that region worth investigating.

5.6 Finding 6 — Cost Structure and Risk Classification

Cost vs Sales scatter analysis confirmed the division-level findings visually. Chocolate transactions cluster tightly with efficient cost-to-sales ratios. Other division products show disproportionately high cost bases relative to their sales values, explaining the 37.67% margin.

Risk flag classification across all 10,194 transactions:

Risk Category	Transaction Count	% of Total	Margin Threshold
Healthy	9,867	96.8%	Above 60%
Reprice	231	2.27%	Between 40-60%
Discontinue Review	96	0.94%	Below 40%

While 96.8% of transactions are healthy, the 327 flagged transactions represent specific products where margin intervention is needed. The Discontinue Review products are predominantly from the Other division, consistent with its structural margin issue.

6. Business Recommendations

Recommendation 1: Protect and Invest in Chocolate Division

Chocolate is not just the best performing division — it is effectively the entire business. With 92.88% revenue share and 67.46% margin, Nassau Candy should treat Chocolate as its core strategic asset. This means prioritising supplier relationships, inventory allocation, and sales effort for Wonka Bar products above all else. Any operational decision that could disrupt Chocolate supply or pricing should be treated as a high-risk business event.

Recommendation 2: Review Wonka Bar Milk Chocolate Pricing

With 8,267 units sold and a 64.92% margin — the lowest among top-performing products — Wonka Bar Milk Chocolate is working hardest for the least return. I identified this product as a margin risk precisely because both conditions were true simultaneously: highest volume AND lowest margin. A targeted price review of 2-3% on this product alone could materially improve profitability given its transaction volume. Nassau Candy should model the revenue impact of a price adjustment before implementation.

Recommendation 3: Conduct Urgent Cost Review of Other Division

The Other division's 37.67% margin is nearly 30 percentage points below Chocolate. This is not a rounding difference — it is a structural gap indicating either excessive supplier costs or significantly underpriced products. Nassau Candy should conduct a line-by-line cost review of Other division products and either renegotiate supplier terms or implement price increases. Products persistently below 40% margin should be formally evaluated for discontinuation.

Recommendation 4: Make a Strategic Decision on Sugar Division

Sugar division has a reasonable margin of 57.69% but generates only \$427.48 in total sales across the entire dataset — less than 0.3% of company revenue. This is not a struggling division; it is an invisible one. Nassau Candy needs to make an explicit strategic choice: either invest in growing Sugar division distribution and awareness, or formally deprioritise it and reallocate those resources to Chocolate. The current passive approach serves neither objective.

Recommendation 5: Reduce Profit Concentration Risk Through Portfolio Development

Five products generating 95% of profit is a business continuity risk. Nassau Candy should actively develop or source at least 3-4 additional products targeting gross margins above 65% to diversify the profit base. This is a medium-term strategic priority. The company's current product mix means a single supplier disruption or market shift in Wonka Bar products could severely damage the entire business.

7. Reflections on the Analytical Process

This was my first complete end-to-end analytics project, and several things genuinely surprised me during the process.

The first was how much the tool matters less than the thinking. Early in the project, I was focused on which tool to use when. By the end, I realised that every tool — Python, SQL, Excel, Power BI, Streamlit — was asking the same analytical questions. The syntax changed, but the logic of groupby, aggregation, and filtering remained constant. Understanding that first in Python made SQL feel familiar, and Excel pivot tables felt like a drag-and-drop version of what I had already done in code.

The second was how honest the data was. I expected the analysis to be more ambiguous — that findings would require interpretation and nuance. Instead, the data told a very clear story once I asked the right questions. Chocolate dominates. Concentration risk is real. High sales does not mean high profit. These findings emerged not from complex modelling but from basic groupby operations applied to well-defined business questions.

The third was the value of cross-validation. Seeing Python, SQL, and Excel all produce the same division margin figures independently gave me a level of confidence in the findings that a single-tool analysis never could. In a professional environment, that cross-validation is what separates a defensible analysis from a guess.

8. Conclusion

This project set out to answer four business questions for Nassau Candy Distributor. All four have been answered with quantitative evidence across multiple analytical tools.

The overarching conclusion is that Nassau Candy operates a highly concentrated, Chocolate-dependent business with strong margin efficiency in its core division but structural weaknesses in Others. The company's immediate priorities should be protecting its Chocolate revenue base, addressing the Other division's margin gap, and making a strategic decision on the Sugar division's future.

From a personal development perspective, this project reinforced that business analytics is fundamentally about asking the right questions before touching any tool. The quality of every output — whether a Python chart, a SQL query, or a Streamlit dashboard — was determined by the clarity of the business question it was trying to answer. That is the lesson I take forward into further projects and into my MSc studies.

9. References and Data Sources

- Dataset: Nassau Candy Distributor Transaction Data (10,194 records, 2024 — provided by Unified Mentor)
- Python 3.12, pandas, matplotlib, plotly — open source analytics libraries
- MySQL 8.0 — Oracle Corporation
- Microsoft Excel — Microsoft Corporation
- Power BI Desktop — Microsoft Corporation
- Streamlit 1.57.0 — Streamlit Inc.
- Unified Mentor Internship Program — Project Brief and Requirements, May 2026